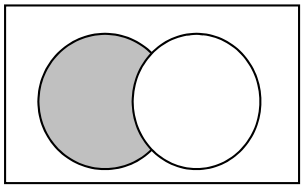
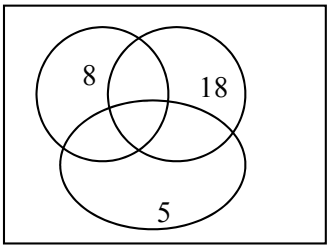
Question	Answer	Marks	Partial Marks
7(a)(i)	111.25	**4**	**M1** for midpoints **soi** (25, 75, 112.5, 137.5, 175) **M1** for $\sum fx$ with x in correct interval including both boundaries **M1** (dep on 2nd **M1**) for $\sum fx \div 20$
7(a)(ii)	2 7 11 17	**2**	**B1** for three correct
7(a)(iii)	$\frac{3}{20}$ oe	**1**	
7(b)	20 6	**2**	**B1** for one correct value or [SF =] 5 or $\frac{1}{5}$ oe
7(c)(i)	5 nfww	**3**	**M2** for $\sum fx \div \sum f = 4.28$ oe or **M1** for $179 + 7x$ oe or $4.28 \times (45 + x)$ oe seen
7(c)(ii)	3	**1**	
7(c)(iii)	4	**1**	
8(a)	–3	**1**	
8(b)	$\frac{12}{11}$ oe	**2**	**M1** for $\dfrac{3}{\frac{3}{x+2}+2}$ soi
8(c)	$64x - 45$ final answer	**2**	**M1** for $8(8x-5)-5$ isw
8(d)	$\frac{x+5}{8}$ oe final answer	**2**	**M1** for a correct first step $y+5=8x$, $\frac{y}{8}=x-\frac{5}{8}$ or $x=8y-5$
8(e)	$\frac{8x^2+11x-13}{x+2}$ final answer	**3**	**M1** for $(8x-5)(x+2)-3$ oe isw **B1** for common denominator $(x+2)$

Question	Answer	Marks	Partial Marks
8(f)(i)	$(8x-5)^2+6 = 19$	**M1**	
	$64x^2-40x-40x+25$	**B1**	
	$64x^2-40x-40x+25+6=19$ oe leading to $16x^2-20x+3=0$	**A1**	with no errors and must show $(8x-5)^2+6 = 19$ with no omissions after this
8(f)(ii)	$\dfrac{[--]20\pm\sqrt{([-]20)^2-4(16)(3)}}{2\times16}$ oe	**2**	**B1** for $\sqrt{([-]20)^2-4(16)(3)}$ or better or **B1** for $\dfrac{[--]20+\sqrt{q}}{2(16)}$ oe or $\dfrac{[--]20-\sqrt{q}}{2(16)}$
	0.17 and 1.08 final ans	**2**	**B1** for each If 0 scored, **SC1** for answer 0.2 and 1.1 or answer – 0.17 and –1.08 or 0.174... and 1.075 to 1.076 seen or 0.17 and 1.08 seen in working
9(a)(i)(a)	$\in$	**1**	
9(a)(i)(b)	$A\cap B$	**1**	
9(a)(ii)	B or A'	**1**	
9(b)		**1**	
9(c)(i)	$3x+7=19$ oe	**M1**	must see 19 and 7
	$3x=19-7$ or better leading to $x = 4$	**A1**	with no errors seen
9(c)(ii)		**2**	**B1** for 2 correct
9(c)(iii)	$\varnothing$ or { }	**1**	
9(c)(iv)	15	**1**	